

Haoran Li

Assistant Professor | Lamar University

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PROFESIONAL EXPERIENCE

Lamar University <i>Assistant Professor, Department of Civil and Environment Engineering</i>	Sept. 2026 – Present Beaumont, TX
Massachusetts Institute of Technology (MIT) <i>Postdoctoral Associate, Department of Civil and Environment Engineering</i>	Feb. 2023 – Sept. 2026 Cambridge, MA
University of Pittsburgh <i>Graduate Student Researcher, Department of Civil and Environment Engineering</i>	Aug. 2018 – Jan. 2023 Pittsburgh, PA
University of Ottawa <i>Graduate Teaching Assistant, Department of Civil Engineering</i>	Sept. 2015 – Apr. 2018 Ottawa, ON Canada

EDUCATION

University of Pittsburgh <i>PhD in Civil Engineering Advisor: Lev Khazanovich</i> Dissertation: Machine Learning Applications for Mechanistic-Empirical Concrete Pavement Design and Analysis (link)	Apr. 2023 Pittsburgh, PA
University of Ottawa <i>MS in Civil Engineering Advisor: Elena Dragomirescu</i> Thesis: Frequency Domain Analysis of Composite Long-Span Cable-Stayed Bridges by Finite Strip Method (link)	Nov. 2017 Ottawa, ON Canada
Southwest Jiaotong University <i>BE in Civil Engineering</i>	Jun. 2015 Chengdu, Sichuan, China

HONORS AND AWARDS

Structural Engineering Institute (SEI) Young Professional Scholarship <i>ASCE Structural Engineering Institute (SEI)</i>	2026
Mark and Roni Magalotti Scholarship in Transportation Engineering <i>University of Pittsburgh</i>	2022
Engineering Graduate Student Organization Travel Grant <i>University of Pittsburgh</i>	2022
2nd Place Winner at Long-Term Pavement Performance (LTPP) Data Analysis Contest <i>The Highway Pavement Committee of the ASCE Transportation & Development Institute (TA&DI) and Federal Highway Administration (FHWA): Paper: Performance of Ohio SPS-2 Sections: Lessons Learned</i>	2020

FUNDED RESEARCH SUPPORT

Market Dynamics of Competing Paving Materials: Quantifying the Cross-Price Effects of Increased Design and Procurement Alternatives	Amount: \$25,000 USD
Role: Principal Investigator Sponsor: American Concrete Pavement Association (ACPA) Foundation Funding Period: January 2026 – December 2026 Description: Applied research project to develop a predictive, state-level modeling framework and public-facing tool that evaluates how procurement strategies, material design flexibility, and inter-material competition influence paving material prices and long-term infrastructure costs. The work builds upon and extends ongoing research efforts at MIT CSHub on material economics and bidding market dynamics.	

RESEARCH EXPERIENCE

Massachusetts Institute of Technology (MIT)

Postdoctoral Associate at Department of Civil and Environmental Engineering
Concrete Sustainability (CS) Hub | Material Systems Laboratory (MSL)

Cambridge, MA

Feb. 2023 – Present

Market Dynamics of Competing Paving Materials: Quantifying the Cross-Price Effects of Increased Design and Procurement Alternatives

Jan. 2026– Present

Sponsor: Concrete Advancement Foundation | **Role:** Lead Investigator

- Develop a forward-looking, state- and national-level model to analyze how procurement strategies (dual bids, alternative design submittals) and material competition (concrete vs. asphalt) influence paving material prices
- Design and propose scenario-based simulations to quantify policy impacts on market dynamics, cost efficiency, and long-term infrastructure investment outcomes
- Outcome includes a predictive tool and policy-relevant framework to guide state DOTs and industry stakeholders in fostering transparent, competitive, and cost-efficient procurement practices

Hydro-Climatic Impacts on Pavement Long-term Performance and Mitigation Strategies

Sponsor: Concrete Sustainability (CS) Hub | **Role:** Lead Investigator

Aug. 2025 - Present

- Investigate impact of extreme precipitation, groundwater fluctuation, and moisture-induced variability to subgrade modulus, load transfer, and pavement performance
- Quantify the coupled effects of hydro-climatic stressors on pavement deterioration and maintenance cycles using mechanistic-empirical and AI-based modeling frameworks
- Develop design and maintenance and asset management optimization strategies to improve pavement resilience under future climate scenarios, such as flooding

Impact of Overloading on Transportation Infrastructure

Jan. 2025 - Present

Sponsor: National Ready Mixed Concrete Association, NRMCA | **Role:** Lead Investigator

- Evaluated the societal and environmental impact of overloaded trucks on the transportation infrastructure, agencies, and truck users from a life-cycle perspective
- Designed a ML-enhanced mechanistic-empirical framework to model long-term pavement behavior under elevated axle loads, validating the feasibility of productivity gains and environmental and financial trade-offs through overloading
- Provided data-driven policy insights supporting performance-based weight regulations, enabling sustainable freight transport solutions amid workforce and environmental constraints

AI-Driven Optimization of Sustainable Pavements

Jan. 2024 - Present

Sponsor: Concrete Advancement Foundation | **Role:** Lead Investigator

- Developed predictive ML and deep reinforcement learning (DRL) models to evaluate pavement performance under diverse design, construction, and maintenance strategies.
- Implemented a Deep Q-Learning (DQN)-based optimization framework that learns optimal life-cycle maintenance and rehabilitation policies under uncertainty, balancing performance, cost, and embodied carbon.
- Integrated LCA and LCCA modules within the AI framework to quantify environmental and fiscal trade-offs across alternative design scenarios.
- Demonstrated that DRL-guided decision policies can reduce greenhouse gas emissions and total costs compared to deterministic or rule-based management strategies, supporting agency-level sustainability planning under budget constraints.

FHWA Climate Challenge Program Projects (collaborate with state DOTs)

Feb. 2023 - Present

- **Indiana Department of Transportation (INDOT)** (sponsor): Led benchmarking of Environmental Product Declarations (EPDs) and LCA for Indiana's concrete pavements to inform low-carbon material, design, and maintenance specifications. | **Role:** Lead Investigator
- **Texas Department of Transportation (TxDOT)** (sponsor): Supported the development of a statewide framework to build TxDOT's capacity for whole-life pavement LCA systems, enabling long-term sustainability evaluation. | **Role:** Lead Investigator
- **Iowa Department of Transportation (IowaDOT)** (sponsor): Applied EPD and LCA to quantify emissions and assess material/design tradeoffs, advancing data-driven sustainable pavement practices in Iowa. | **Role:** Lead Investigator

Development of a Streamlined Pavement Life Cycle Assessment (LCA) and Life Cycle Cost Analysis (LCCA) Tool

Feb. 2023 - Present

Sponsor: Concrete Advancement Foundation | **Role:** Lead Investigator

- Built surrogate ML models using adaptive data sampling, enabling long-term pavement performance forecasting using mechanistic-empirical pavement design framework
- Developed a streamlined Life Cycle Assessment (LCA) framework and a software tool for both asphalt and concrete pavements (<https://pavementlca.mit.edu/>)
- Informed robust decisions to stakeholders at early stages by addressing uncertainty in life cycle impacts

University of Pittsburgh

Graduate Student Researcher

Department of Civil and Environmental Engineering

Pittsburgh, PA

Aug. 2018 - Jan. 2023

Development of A Novel Adaptive Data Sampling Method for Surrogate Modeling Using Machine Learning

Sponsor: University of Pittsburgh Anthony Gill Chair | **Role:** Lead Investigator

Sept. 2021 - Jan. 2023

- Developed a new adaptive data sampling method to improve the efficiency and accuracy of surrogate models for rigid pavement design across high-dimensional parameter spaces
- Applied the Scalable Adaptive Sampling (SAS) approach to both benchmark problems and real-world pavement design scenarios, achieving an order of magnitude lower error in model predictions
- Demonstrated that the proposed SAS approach reduces required sample sizes by up to 95% compared to conventional methods while maintaining or improving predictive performance
- Pending patent application: PCT/US2024/010695 (WO2024151535A1): see IP1 in CV

Climate Risk Assessment and Design of Heat-Resilient Concrete Pavements

Jan. 2022 - Jan. 2023

Sponsor: University of Pittsburgh Anthony Gill Chair | **Role:** Lead Investigator

- Developed simulation models incorporating future climate projections and evaluated performance uncertainties using probabilistic analysis
- Evaluated the impact of climate change and urban heat islands (UHI) on long-term concrete pavement performance using mechanistic-empirical simulations in diverse U.S. climates
- Quantified the deterioration change of concrete pavements under rising daily temperature extremes, supporting the development of climate-adaptive pavement design strategies

Faulting Model Improvements for Mechanistic-Empirical Pavement Design Guide

Jun. 2019 - Aug. 2021

Sponsor: The Pennsylvania Department of Transportation, PennDOT | **Role:** Lead Investigator

- Developed a novel methodology for locally calibrating the Pavement ME faulting model using high-resolution Pavement Management System (PMS) data to support high-reliability pavement design
- Addressed challenges of limited and censored field data by leveraging large-scale PMS datasets, enabling more accurate predictions of joint faulting at higher reliability levels
- Validated the calibrated model using Pennsylvania pavement data and demonstrated its applicability for broader mechanistic-empirical model refinement across state and national networks

Development of Simplified Mechanistic-Empirical (M-E) Design Tool for Pennsylvania Rigid Pavements

Sponsor: Impactful Resilient Infrastructure Science and Engineering (IRISE) Consortium | **Role:** Lead Investigator

Jan. 2019 – Dec. 2020

- Streamlined pavement design by identifying and utilizing only the most influential inputs for local conditions, significantly reducing computational effort and risk of error
- Developed PittRigid ME, a localized and simplified mechanistic-empirical (M-E) design tool tailored to Pennsylvania concrete pavements, reducing input complexity while maintaining predictive accuracy (<https://software.pavements.pitt.edu/PittRigid>)
- Enabled faster, more reliable pavement design and performance predictions (cracking, faulting, IRI) aligned with AASHTO MEPDG principles, with potential scalability to other state DOTs

Evaluation of Long-Term Impacts of Early Opening of Concrete Pavements

Sep. 2018 – Dec. 2020

Sponsor: Minnesota Department of Transportation, MNDOT | **Role:** Lead Investigator

- Investigated the short-term response and long-term performance of early-opened concrete pavements using MnROAD field data, identifying optimal strength thresholds for safe traffic opening without compromising durability
- Conducted lab and field experiments on concrete beams, cylinders, and pavement slabs regarding early-age concrete strength and maturity
- Performed mechanistic analyses with ISLAB2000 to assess the impact of load location and thermal gradients on early-age pavement stress and potential damage
- Contributed to the development of a web-based decision tool for determining site-specific early opening timing, enabling faster project delivery with maintained long-term pavement integrity (<https://software.pavements.pitt.edu/EOP-National>)

University of Ottawa

Ottawa, ON Canada

Research Assistant at Department of Civil Engineering | **Role:** Lead Investigator

Aug. 2015 – Jun. 2017

Frequency Domain Analysis of Composite Long-span Cable-stayed Bridges by Finite Strip Method

- Developed and applied a laminate finite strip method (LFSM) to model the static and dynamic behavior of long-span composite cable-stayed bridges using fiber-reinforced polymer (FRP) decks
- Conducted time history, frequency domain, and free vibration analyses of the Kap Shui Mun Bridge, validating results against finite element models
- Demonstrated LFSM's computational efficiency and accuracy and its capability to perform stable time and seismic response analyses in cases where conventional finite element approaches face convergence issues

TEACHING EXPERIENCE AND PEDAGOGY TRAINING

Massachusetts Institute of Technology (MIT)

Cambridge, MA

Faculty Mentor / Peer Coaching Contributor (Communication Fellow)

Mar. 2025 – Present

MIT Civil & Environmental Engineering Communication Lab

- Provided individualized coaching for undergraduate, graduate, and postdoctoral researchers on CV and résumé development, tailoring application materials, and presentation of materials with a discipline-specific focus
- Guided peers in preparing manuscripts for publication, designing effective scientific posters, and refining oral presentations to suit diverse academic and industry audiences.
- Advised on competitive fellowship, grant, and proposal submissions by offering structural feedback, clarity improvements, and strategic alignment of technical content

Teaching Fellow - Kaufman Teaching Certificate Program (KTCP)

Sept. 2023 – Dec. 2023

- Completed MIT's competitive teaching development program, focused on pedagogical best practices, inclusive teaching, and active learning strategies
- Developed a standardized syllabus for a future graduate-level course titled "Design and Analysis of Sustainable Road Pavement," integrating learning objectives, course outcomes, assessment methods, and weekly modules
- Designed and delivered two microteaching sessions with peer and instructor feedback, demonstrating effective teaching techniques aligned with course design principles

University of Pittsburgh

Pittsburgh, PA

Summer Teaching Fellow

June 2022 – July 2022

Office of the Provost & University Center for Teaching and Learning

- Evaluated the long-term impacts of instructional innovations on student learning outcomes across diverse educational settings, such as guest lectures, workshops, etc.
- Gained in-depth knowledge of pedagogical strategies and evidence-based practices in higher education teaching
- Participated in professional development workshops focused on academic career preparation and inclusive instructional design

Graduate Teaching Assistant

Sept. 2018 – Dec. 2022

Department of Civil and Environmental Engineering

CEE 2714 Pavement Design and Analysis; CEE 3714 Advanced Pavement Design and Analysis

- Supported instruction of undergraduate and graduate-level pavement design courses by guiding students through complex technical concepts and design methodologies
- Designed and prepared assignments, term projects, and solution sets to reinforce course learning objectives
- Assessed student performance through grading of assignments, quizzes, and exams, and provided individualized, constructive feedback to support academic development

University of Ottawa

Ottawa, ON, Canada

Graduate Teaching Assistant

Sept. 2015 – May. 2017

Department of Civil Engineering

CVG 2149 Civil Engineering Mechanics; CVG 2549 Mécanique de génie civil; CVG 5154 Random Vibrations

- Delivered lectures, tutorials, and lab demonstrations to support student learning in both undergraduate and graduate engineering courses
- Designed, supervised, and evaluated lab experiments, guiding students through mechanical testing procedures and analysis
- Assessed coursework by preparing and grading assignments, quizzes, exams, and term projects, while providing feedback to enhance student understanding

PUBLICATIONS

Peer-Reviewed Journal Articles

- (J1) **Li, H.**, Azarijafari, H., Kirchain, R. (2026). Sustainable Urban Concrete Delivery: AI and LCA-Driven Insights into Fully-Loaded Ready-Mixed Concrete Trucks and Their Societal Impacts. **Discover Cities**. (Under Review).
- (J2) **Li, H.**, Sen, S., Khazanovich, L. (2026). Surrogate Modeling of Transverse Joint Faulting Using Artificial Neural Networks for Mechanistic-Empirical Design of Concrete Pavements. **Journal of Transportation Engineering, Part B: Pavements**. (Under Review).
- (J3) Shu, Z, **Li, H.**, Sun, L., Yan, Y. (2026). An enhanced HMA-FM model for predicting the fatigue life of asphalt mixtures based on physics-informed deep learning. **Road Materials and Pavement Design**. (Under Review).

- (J4) **Li, H.**, Sen, S., Khazanovich, L. (2026). Adaptive Sampling and Artificial Neural Networks for Surrogate Modelling in High-Dimensional Mechanistic-Empirical Rigid Pavement Design and Analysis. **Road Materials and Pavement Design**. doi: [10.1080/14680629.2026.2696984](https://doi.org/10.1080/14680629.2026.2696984)
- (J5) Fang, M., Han, L., Wang, L., Xiao, H., Amédée, S., **Li, H.**, Roesler, J. (2026). Advancements and Challenges in Precast Concrete Pavement Joint Technology: A Review of Joint Types, Materials, and Construction Methods, **Journal of Traffic and Transportation Engineering (English Edition)**. (In press).
- (J6) **Li, H.**, Zhang, M., Azarijafari, H., Kirchain, R. (2025). Advancing Smart and Sustainable Road Construction and Maintenance through Life-Cycle Optimization of Concrete Pavements. **Intelligent Transportation Infrastructure**. 4. doi: [10.1093/iti/liaf029](https://doi.org/10.1093/iti/liaf029)
- (J7) **Li, H.**, Azarijafari, H., Kirchain, R., Santos, J., Khazanovich, L. (2024). Surrogate Modelling of Surface Roughness for Asphalt Pavements Using Artificial Neural Networks: A Mechanistic-Empirical Approach. **International Journal of Pavement Engineering**. 25 (1), 2434909, doi: [10.1080/10298436.2024.2434909](https://doi.org/10.1080/10298436.2024.2434909).
- (J8) **Li, H.**, Zhang, M., Liu, H., Azarijafari, H., Kirchain, R. (2024). Development of A Streamlined Framework for Probabilistic and Comparative Life Cycle Assessment of Pavements. **Resources, Conservation and Recycling**. 209, 107802. doi: [10.1016/j.resconrec.2024.107802](https://doi.org/10.1016/j.resconrec.2024.107802)
- (J9) **Li, H.**, Sen, S., Khazanovich, L. (2024). A Scalable Adaptive Sampling for Surrogate Modeling Approach of Rigid Pavements Using Machine Learning. **Results in Engineering**. 23, 102483. doi: [10.1016/j.rineng.2024.102483](https://doi.org/10.1016/j.rineng.2024.102483)
- (J10) De Salles, L., **Li, H.**, Khazanovich, L. (2024). Local High Reliability Calibration of Faulting Model using Pavement Management Data. **Transportation Research Record**. 1-16. doi: [10.1177/03611981241253596](https://doi.org/10.1177/03611981241253596)
- (J11) **Li, H.**, & Khazanovich, L. (2022). Multi-gene genetic programming extension of AASHTO ME for design of low-volume concrete pavements. **Journal of Road Engineering**. 2(3), 252-266. doi: [10.1016/j.jreng.2022.08.002](https://doi.org/10.1016/j.jreng.2022.08.002)
- (J12) Sen, S., **Li, H.**, Khazanovich, L. (2022). Effect of climate change and urban heat islands on the deterioration of concrete roads. **Results in Engineering**. Vol. 16, p. 100736, doi: [10.1016/j.rineng.2022.100736](https://doi.org/10.1016/j.rineng.2022.100736)
- (J13) **Li, H.**, & Khazanovich, L. (2021). PITTRIGID ME: Simplified Mechanistic-Empirical Design Tool for Pennsylvania Rigid Pavements Design and Analysis. **Journal of Transportation Engineering, Part B: Pavements**. 147(4), 04021052. doi: [10.1061/JPEODX.0000307](https://doi.org/10.1061/JPEODX.0000307)

Peer-Reviewed Conference Papers

- (C1) **Li, H.**, Zhang, M., Azarijafari H., Kirchain, R. (2024). Life Cycle Assessment of Continuously Reinforced Concrete Pavement Overlay on Concrete Road Rehabilitation: Insights from Two Case Studies. **6th International Symposium on Pavement, Roadway, and Bridge Life Cycle Assessment**, Arlington, VA, USA. doi: [10.1007/978-3-031-61585-6_23](https://doi.org/10.1007/978-3-031-61585-6_23)
- (C2) Zhang, M., **Li, H.**, Azarijafari H., Kirchain, R. (2024). Optimizing Network-Level Pavement Treatment for Greenhouse Gas Emission Reduction. **6th International Symposium on Pavement, Roadway, and Bridge Life Cycle Assessment**, Arlington, VA, USA. doi: [10.1007/978-3-031-61585-6_24](https://doi.org/10.1007/978-3-031-61585-6_24)
- (C3) **Li, H.**, Sen, S., and Khazanovich, L. (2024). Artificial Neural Networks for Mechanistic-Empirical Concrete Pavement Design: An Efficient Development Approach. **13th International Conference on Concrete Pavements**. Minneapolis, Minnesota, USA
- (C4) **Li, H.**, Levin, J.R., Azarijafari, H., Gregory, J., Kirchain, R., Mack, J.W., Zollinger, C.J., and Mahdi, M. (2024). Lifecycle Assessment and Sensitivity Analysis of Roller Compacted Concrete, Jointed Plain Concrete, and Hot Mix Asphalt Pavements. **13th International Conference on Concrete Pavements**. Minneapolis, Minnesota, USA
- (C5) **Li, H.**, & Khazanovich, L. (2023). Machine Learning-Based Symbolic Expressions for Mechanistic-Empirical Rigid Pavement Design and Performance Prediction. **Advances in Materials and Pavement Performance Prediction (AM3P)**, Hong Kong.

Reports

- (R1) Khazanovich, L., Kosar, K., **Li, H.** (2021). Evaluation of Long-Term Impacts of Early Opening of Concrete Pavements. NARA202111. **Minnesota Department of Transportation**. St. Paul, Minnesota ([Link](#))
- (R2) Khazanovich, L. & **Li, H.** (2020). Development of Simplified Mechanistic-Empirical Design Tool for Pennsylvania Rigid Pavements. IRISE-20-P20-01-01. **PITT IRISE Swanson School of Engineering**. University of Pittsburgh. Pittsburgh, PA ([Link](#))

Research Briefs

- (RB1) **Li, H.**, Azarijafari H., and Kirchain, R. (2025). Life Cycle Impacts of Fully-Loaded Ready Mixed Concrete (RMC) Trucks. **MIT Concrete Sustainability Hub**. Volume 2025. (<https://cshub.mit.edu/files/2025/07/Fully-Loaded-RMC-Trucks.pdf>)

Intellectual Properties

- (IP1) **Li, H.**, Sen, S., and Khazanovich, L. *Scalable Adaptive Sampling Methods for Surrogate Modeling Using Machine Learning*. U.S. Pat. Ser. No. PCT/US24/10695. January 08, 2024. (<https://patents.google.com/patent/WO2024151535A1/en>)

PRESENTATIONS AND TALKS

Conference Presentations

- (P1) **Li, H.**, Azarijafari H., Swei O., Kirchain, R. (2025). From Pavement Strain to Policy Gain: A Life-Cycle Thinking and Multi-Objective Approach to Truck Overloading and Infrastructure Management. **105th Transportation Research Board Annual Meeting**. Washington D.C. January 2026.
- (P2) Zhang, M., **Li, H.**, Azarijafari, H., Kirchain, R., Benchmarking and Reducing GHG Emissions in Pavement Networks, **104th Transportation Research Board Annual Meeting**. Washington D.C. January 2025
- (P3) **Li, H.**, Zhang, M., Azarijafari, H., Kirchain, R., Optimizing Design and Maintenance: A Pathway to Decarbonizing Concrete Pavements' Life Cycle, **104th Transportation Research Board Annual Meeting**. Washington D.C. January 2025
- (P4) **Li, H.** Lifecycle Assessment and Sensitivity Analysis of Roller Compacted Concrete, Jointed Plain Concrete, and Hot Mix Asphalt Pavements. **104th Transportation Research Board Annual Meeting**. Washington D.C. January 2025
- (P5) Zhang, M., **Li, H.**, Liu, H., Azarijafari, H., Kirchain, R., Comprehensive GHG Inventory and Avoided Emissions Analysis in Pavement Sectors: A Novel Framework. **103rd Transportation Research Board Annual Meeting**. Washington D.C. January 2024
- (P6) **Li, H.**, Zhang, M., Azarijafari, Kirchain, R., Development of a Streamlined Framework for Probabilistic and Comparative Life Cycle Assessment of Concrete Pavements, **ACI Concrete Convention**. Boston, MA October 2023
- (P7) **Li, H.**, Sen, S., and Khazanovich L. Development of Artificial Neural Networks for Mechanistic-Empirical Concrete Pavement Design Using A New Adaptive Sampling Method. **Keynote Presentation. 2nd International Data Science for Pavements Symposium**, McLean, VA. March 2023
- (P8) Sen, S., **Li, H.**, and Khazanovich L. Equivalent Single Axle Loads in Rigid Pavements: A Mechanistic-Empirical Perspective. **102nd Transportation Research Board Annual Meeting**. Washington D.C. January 2023
- (P9) **Li, H.**, & Khazanovich, L. Multi-Gene Genetic Programming Extension of AASHTO M-E for Design of Low-Volume Concrete Pavements. **101st Transportation Research Board Annual Meeting**. Washington D.C. January 2022
- (P10) **Li, H.**, & Khazanovich, L. Artificial Neural Networks Based Assessment Tool and Design Alternative for AASHTO-ME Concrete Pavements Design. **1st International Data Science for Pavements Symposium**, McLean, VA. March 2022

Public Webinars

- (W1) **Li, H.** *From Designs to Decisions: Using MIT's Online Pavement LCA/LCCA Tool for Low-Carbon, Cost-Effective Choices (Series 2/2)*. MIT Concrete Sustainability Hub. Jun. 25, 2026. https://youtu.be/y5fKXmI6mGo?si=oi0KoDsLj_7BXj4l
- (W2) **Li, H.** *From Designs to Decisions: Using MIT's Online Pavement LCA/LCCA Tool for Low-Carbon, Cost-Effective Choices (Series 1/2)*. MIT Concrete Sustainability Hub. Jan. 29, 2026. <https://www.youtube.com/watch?v=tRdhmRMwSVc>
- (W3) **Li, H.** *Fuller Loads, Lighter Footprint: Rethinking RMC Truck Efficiency Through Life Cycle Analysis*. MIT Concrete Sustainability Hub. Sept. 25, 2025. <https://www.youtube.com/watch?v=mJTHU8h-wol>
- (W4) **Li, H.** *Development of A Streamlined Pavement Life Cycle Assessment Tool*. MIT Concrete Sustainability Hub. Feb. 27, 2025. https://www.youtube.com/watch?v=H_PqaY-5f04
- (W5) **Li, H.** *Analyses from MIT CSHub Streamlined Pavement LCA Tool*. MIT Concrete Sustainability Hub. Dec. 14, 2023. <https://www.youtube.com/watch?v=NvOlpKinBYI>

Invited Talks

- (IT1) **Li, H.** Graduate Career Design Workshop Series: *Create or Update Your LinkedIn Profile*. Office of the Provost, University of Pittsburgh, Pittsburgh, PA. Oct. 18, 2022.

PROFESSIONAL SERVICE

Committee Member of Transportation Research Board (TRB) 2025 - Present

- AKP20 - Standing Committee on Design and Rehabilitation of Concrete Pavements
- AKN14 - Standing Committee on Concrete Pavement Design and Construction

MIT Department of Civil and Environmental Engineering

Communication Fellow

2025 - 2026

Postdoctoral Researcher Committee Member

2024 - 2026

Road Materials and Pavement Design (Journal)

Member of Editorial Board

2025 - Present

MIT Postdoctoral Association (PDA)

IT & Information Committee Co-Chair

2023 - 2024

Peer-Review Articles for:

2018 - Present

- Transportation Research Board Annual Meeting (TRBAM)
- Transportation Research Record (TRR)
- Results in Engineering (RINENG)
- Engineering Applications of Artificial Intelligence (EAAI)
- Resources, Conservation and Recycling (RCR)
- International Journal of Sustainable Transportation (IJST)
- Multidiscipline Modeling in Materials and Structures (MMMS)
- International Journal of Pavement Engineering (IJPE)
- International Journal of Pavement Research and Technology
- Journal of Transportation Engineering, Part B: Pavements
- Road Materials and Pavement Design (RMPD)

MEDIA COVERAGE

"3 Questions: Making the most of limited data to boost pavement performance" - MIT News May 14, 2025

- Featured interview on developing advanced predictive models and improving pavement's life-cycle performance through machine learning ([link](#))

"New framework empowers pavement life-cycle decision-making while reducing data collection burden"

- MIT News

Aug. 9, 2024

- Coverage of **Haoran Li's** work on the streamlined pavement LCA tool and its adoption in state agency workflows ([link](#))

"Climate Change Creates Complications for Concrete" – University of Pittsburgh Swanson School of Engineering News

Nov. 14, 2022

- Coverage of **Haoran Li's** work using mechanistic-empirical modeling to reveal how extreme temperature accelerates concrete pavement deterioration under climate change ([link](#))

"Student Research Spotlight – August 2022" – International Society for Concrete Pavements

Aug. 24, 2022

- Coverage of **Haoran Li's** work on the ML-enhanced mechanistic-empirical design and analysis of pavements ([link](#))

PROFESSIONAL AFFILIATIONS

- | | |
|---|--------------|
| • American Society of Civil Engineers (ASCE) | 2018-Present |
| • International Society for Concrete Pavements (ISCP) | 2018-Present |
| • Canadian Society for Civil Engineering (CSCE) | 2015-Present |

PROFESSIONAL TRAINING AND CERTIFICATE

Massachusetts Institute of Technology (MIT) Cambridge, MA

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| • Leadership Management in Action Program (L-MAP) | Jan. 2026 |
| • Professional Development Certificate in Research Mentoring Certificate | Jul. 2025 |
| • Grant Writing Training Professional Development Certificate | Jan. 2025 |
| • Life Cycle Assessment: Quantifying Environmental Impacts Certificate | Jun. 2024 |
| • Kaufman Teaching Certificate Program (KTCP) | Jan. 2024 |

Harvard University Cambridge, MA

- | | |
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| • Certificate Coding Certificate | Jan. 2026 |
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University of Pittsburgh Pittsburgh, PA

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|---|-----------|
| • Pitt-CIRTL Associate Level Certification in STEM Teaching | Jan. 2019 |
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University of Ottawa Ottawa, ON Canada

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| • Dry Lab Risk Management Certificate | May 2016 |
| • Workplace Hazardous Materials Information System (WHMIS) Certificate | May 2016 |

Coursera Online

- | | |
|---|------|
| • Neural Networks and Deep Learning (credential) | 2022 |
| • Applied Machine Learning in Python (credential) | 2021 |

LIST OF MENTEES

Massachusetts Institute of Technology (MIT) Cambridge, MA

Rudiba Laiba (MEng student in Electrical Engineering and Computer Scienc) Sep. 2025 – Present

- Development of a forward-looking, state- and national-level model to evaluate material competition and procurement strategies in pavement markets.
- Mentored in econometric modeling, scenario-based simulation design, and data-driven policy analysis to quantify cross-price effects and market dynamics.

An Dinh (undergraduate student in Computer Science) Jan. 2025 – Aug. 2025

- Implementation of Pavement LCCA Model into the online software tool, including the front-end UI/UX and back-end modeling.
- Mentored in full-stack software development best practices, guiding integration of engineering models into user-friendly online platforms

Tissany Chen (undergraduate student in Mechanical Engineering) May 2023 – Aug. 2024

- Applications and Analysis of the Streamlined Pavement Life Cycle Assessment Tool
- Provided hands-on training in environmental impact assessment methods and data analysis for infrastructure sustainability

Varsha Vaidyanath (MS student in Technology and Policy Program)

Sept. 2023 - May 2024

- **Master Thesis:** *Benchmarking Pavement Environmental Performance Using Data-Driven Methods*
- Supervised thesis development, offering strategic guidance on methodological design, statistical analysis, and policy implications

Hannah Han (undergraduate in Artificial Intelligence and Decision Making)

Sept. 2023 - May 2024

- Development of a Software Tool for Pavement Life Cycle Assessment
- Advised on AI-driven design principles, ensuring robust modeling integration with intuitive decision-support interfaces.

Maeve Chen (undergraduate in Civil and Environmental Engineering)

Jan. 2024

- Applications and Analysis of the Streamlined Pavement Life Cycle Assessment Tool
- Mentored in applying life cycle assessment techniques to civil infrastructure, emphasizing data interpretation and model validation

PROFESSIONAL REFERENCES

Lev Khazanovich, Ph.D. (PhD Advisor)

Anthony Gill Chair Professor
Department of Civil and Environmental Engineering
University of Pittsburgh
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